



MINING DEALER BEST PRACTICE

R1300G Final Drive Bolt Spinning Prevention Fixture

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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February 2016
1511-4.03-1402

1.0 Introduction

Borusan Makina Kazakhstan (BMK) has designed a fixture that prevents the bolts from spinning in R1300G final drives during their multiple tightening.

Previously the BMK mining project teams constantly faced the problem of the bolts spinning in R1300G final drives during tightening. To remedy this, they welded those bolts to the surface of the final drive. When this final drive came to the CRC for overhaul, the CRC technicians had to clean the surface of the final drive from welding. Thus, the spinning bolts have become the common challenge both for the mining and for the CRC people.

That's why the CRC has created the fixture that allows everybody to forget about this concern. The application of this fixture enables the mining technicians to easily tighten those bolts during the operation and has saved the CRC people from cleaning the surface of the final drive from welding during the overhaul.

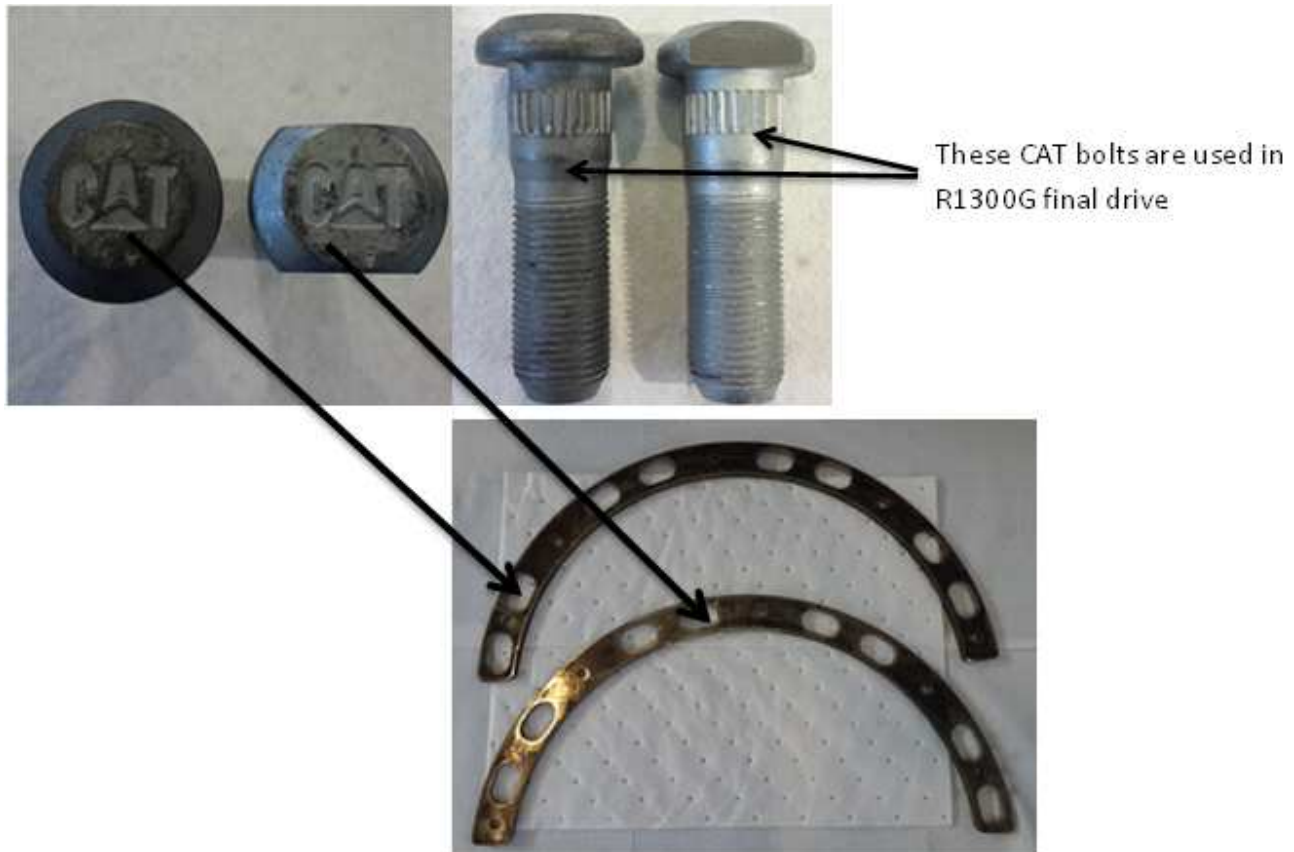
At time of publishing, Borusan Makina Kazakhstan has reworked 16 hubs and have not experienced any failures in the fields for nearly a year.

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2.0 Best Practice Description

This fixture consists of 2 x Locking semi-rings. These semi-rings have the holes for the bolt heads that lock the bolts. They are bolted to the final drive surface with smaller bolts (10 pcs).



Semi-rings with locking holes for R1300G Final Drive Bolts (P/N 308-9259).

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3.0 Implementation Steps

R1300G Final Drive Bolt Spinning Prevention Fixture is in use.

- 1) Bolt 2 x Locking Semi-rings to the surface of the final drive.
- 2) Install CAT Bolts into to the final drive.

Note: Loctite 640 is applied to the smaller bolts.



R1300G Final Drive Bolt Spinning Prevention Fixture is in use

4.0 Benefits

- Prevention of the final drive bolts from spinning during their tightening.
- There is no need to clean the surface of the final drive from welding during the overhaul in the CRC.

5.0 Resources Required

- Machine Shop.
- Cat Bolts – P/N 308-9259

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Editor's note: 308-9259 bolt was a service replacement bolt with an anti-corrosion coating that was considered to be the cause of the slipping/spinning. It has been reverted back to use the original 1V-3323 bolts.

6.0 Supporting Attachments / References

The semi-ring's drawing is attached.

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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